Comparative analysis of the *Eulemur* vocal repertoire: a dynamic time warping approach.

Authors: Marco Gamba[a], Olivier Friard[a], Isidoro Riondato[a], Roberta Righini[a], Camilla Colombo[a], Longondraza Miaretsoa[b], Valeria Torti[a], Bakri Nadhurou[a], Cristina Giacoma[a].

[a] Department of Life Sciences and Systems Biology, University of Torino, Italy.
[b] Faculté de Sciences, University of Mahajanga, Madagascar.

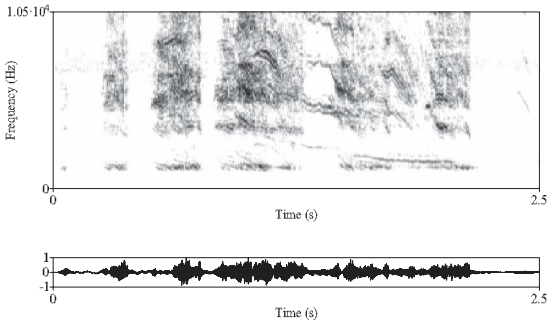
Corresponding author: Marco Gamba, Department of Life Sciences and Systems Biology, University of Torino, Via Accademia Albertina 13, 10123 Torino, Italy.
Tel. +390116704560
Fax. +390116704508
Email: marco.gamba@unito.it

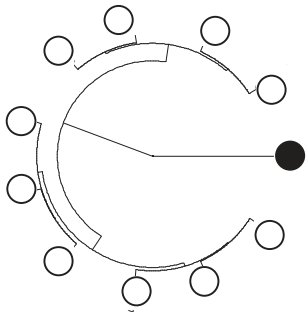
Appendix S6. ***We reported a large spectrogram of each vocalization chosen as exemplar during cluster analysis, its waveform, and a schematic representation of the polar dendrogram in Figs. 1-7 showing cluster position.***

Eulemur fulvus

a

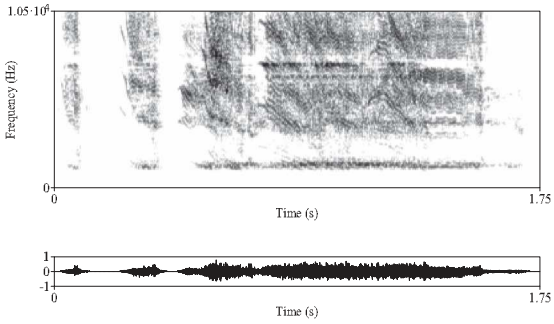
Territorial Advertisement Call

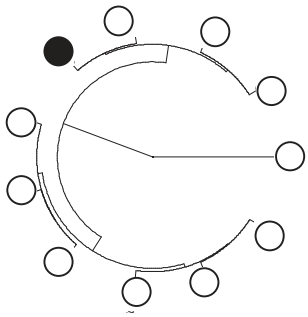




b

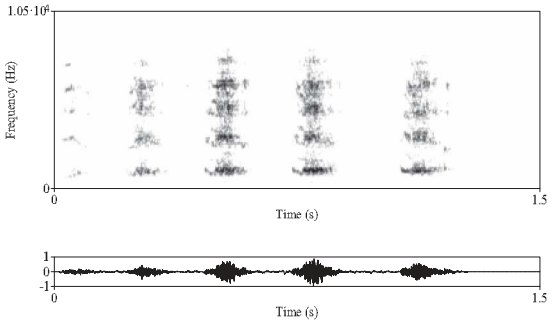
Territorial Advertisement Call

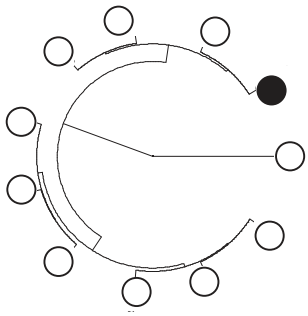




c

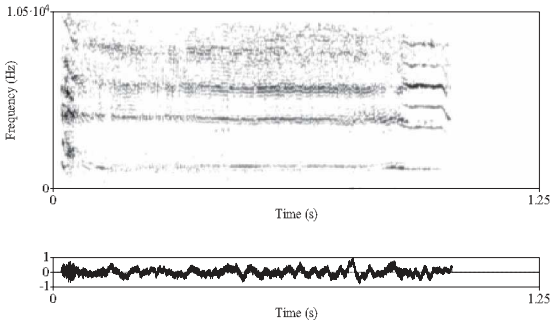
Group Cohesion Call

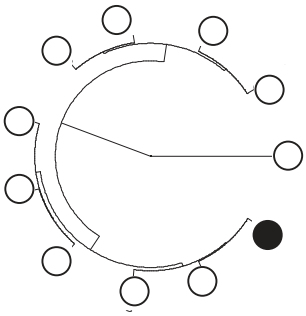




d

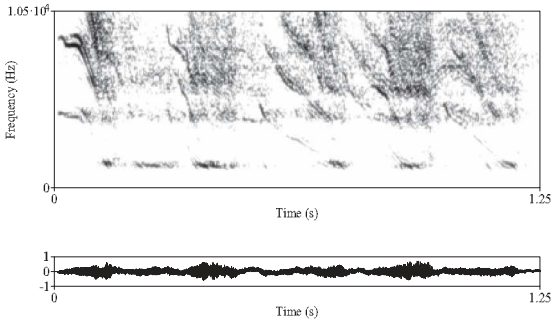
Long Grunt-Tonal Call

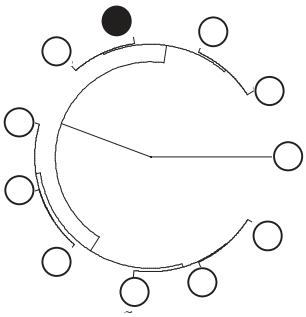




e

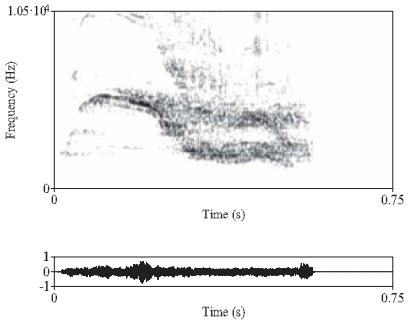
Territorial Advertisement Call

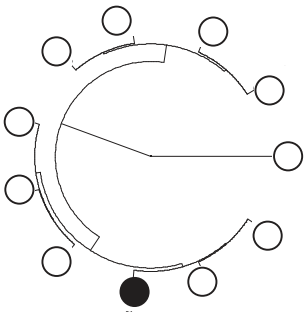




f

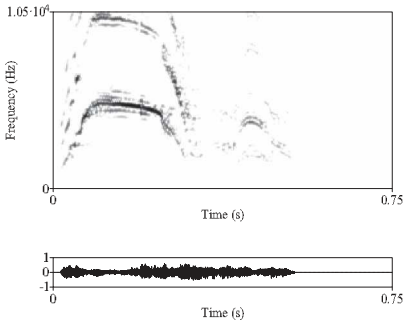
Scream

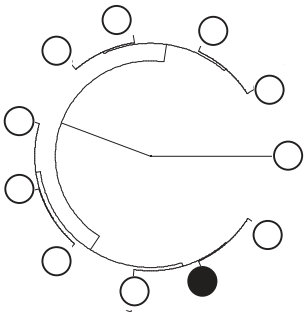




g

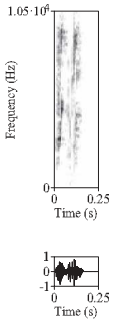
Scream

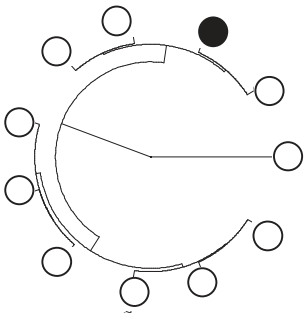




h

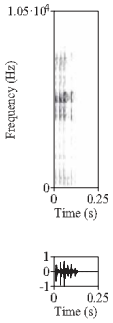
Chatter

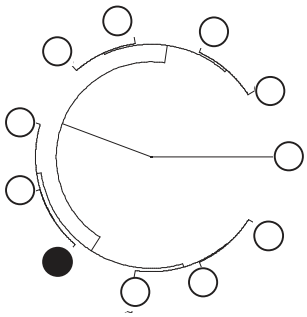




i

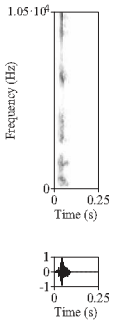
Grunt





j

Grunt



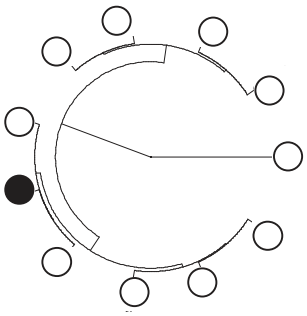
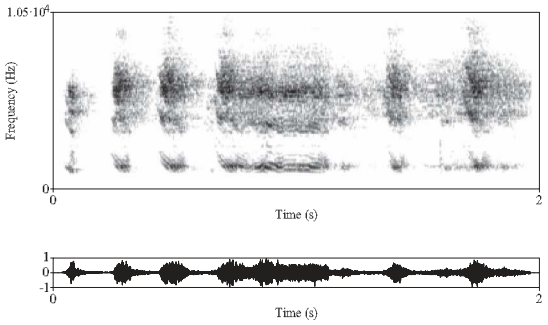


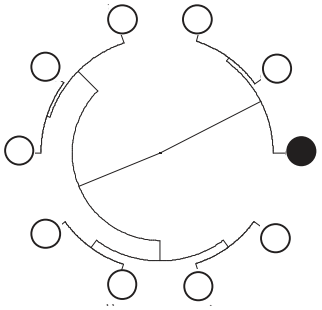
Fig. 16 Spectrograms and waveforms of the exemplars chosen from the vocalization of *Eulemur fulvus* during the cluster analysis: Territorial Advertisement Call (a, refers to Table SII in ESM Appendix S5; Cluster ID 6; window length: 0.01 s); Territorial Advertisement call (b, refers to Table SII in ESM Appendix S5; Cluster ID 10; window length: 0.01 s); Group Cohesion Call (c, refers to Tab. S2 in Online Material 3 Appendix S5; Cluster ID 5; window length: 0.01 s); Long Grunt-Tonal Call (d, refers to Tab. S2 in Online Material 3 Appendix S5; Cluster ID 7; window length: 0.01 s); Territorial Advertisement Call (e, refers to Tab. S2 in Online Material 3 Appendix S5; Cluster ID 3; window length: 0.01 s); Scream (f, refers to Tab. S2 in Online Material 3 Appendix S5; Cluster ID 11; window length: 0.01 s); Scream (g, refers to Table SII in ESM Appendix S5; Cluster ID 1; window length: 0.01 s); Chatter (h, refers to Table SII in ESM Appendix S5; Cluster ID 9; window length: 0.01 s); Grunt (i, refers to Table SII in ESM Appendix S5; Cluster ID 4; window length: 0.01 s); Grunt (j, refers to Table SII in ESM Appendix S5; Cluster ID 2; window length: 0.005 s). A schematic representation of the polar diagram (Fig. 1) shows cluster and spectrogram position in the original figure.

Eulemur rufus

a

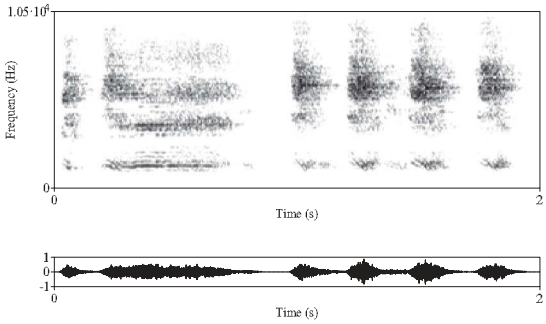
Territorial Advertisement Call

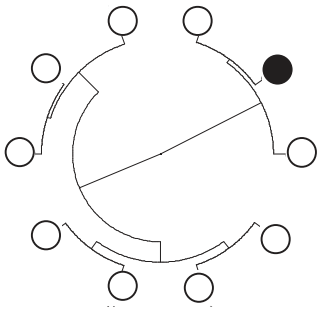




b

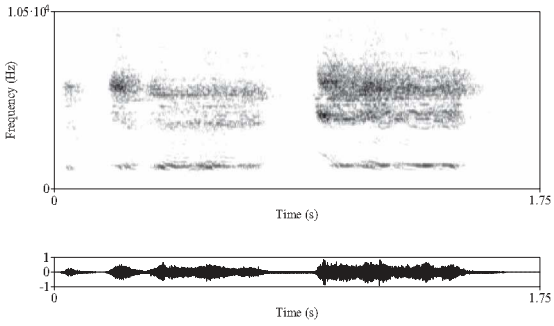
Territorial Advertisement Call

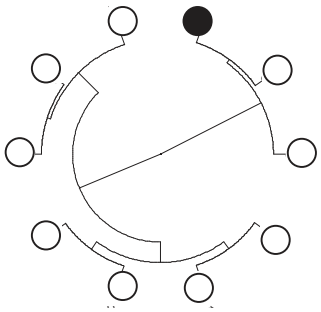




c

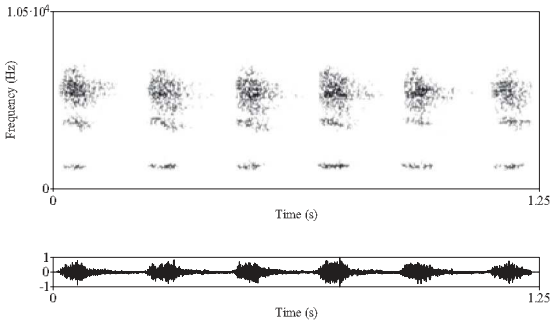
Territorial Advertisement Call

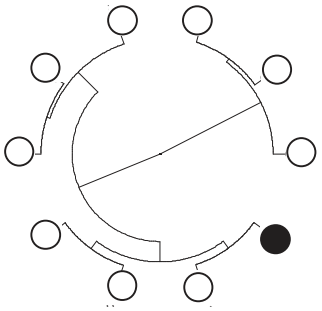




d

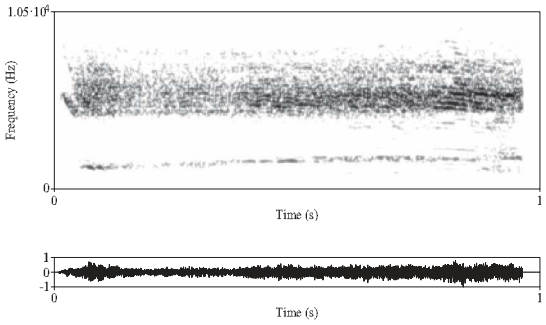
Group Cohesion Call

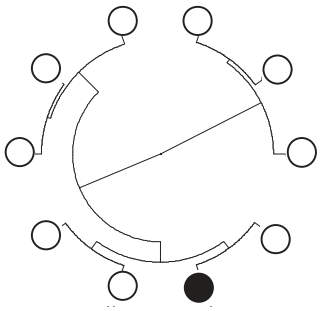




e

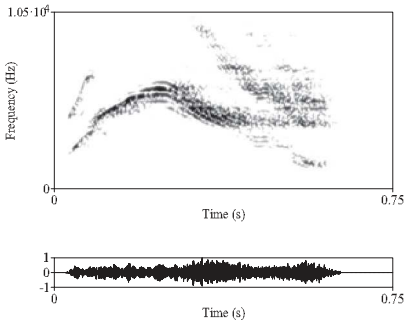
Alarm Call

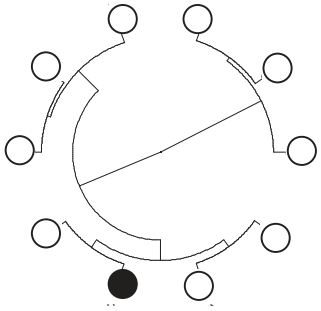




f

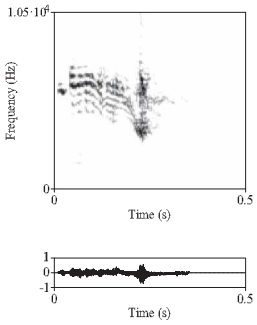
Scream

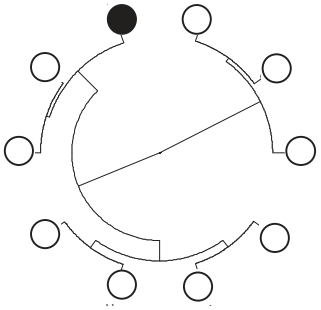




g

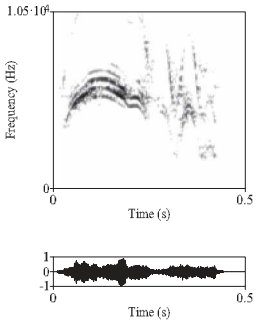
Scream

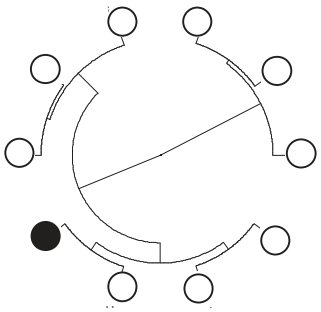




h

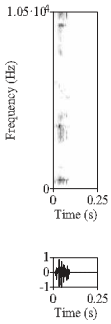
Scream

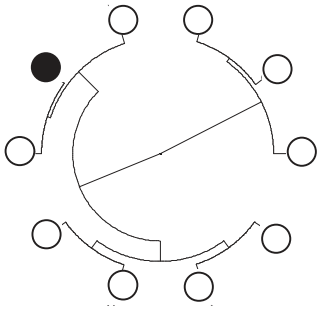




i

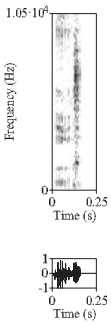
Grunt





j

Grunted Hoot



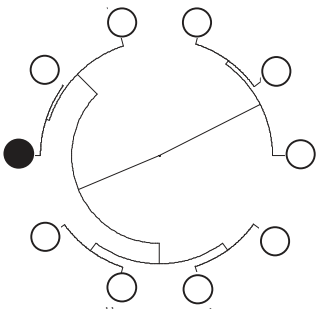
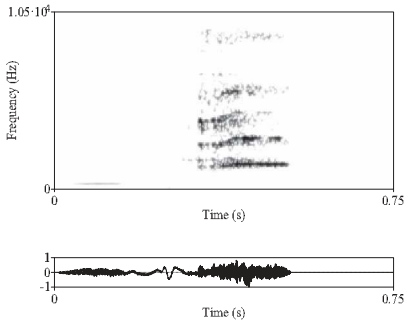


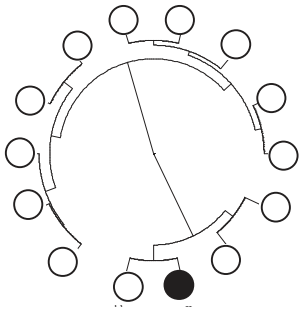
Fig. 17 Spectrograms and waveforms of the exemplars chosen from the vocalization of *Eulemur rufus* during the cluster analysis: Territorial Advertisement call (a, refers to Table SIII in ESM Appendix S5; Cluster ID 9; window length: 0.01 s); Territorial Advertisement call (b, refers to Table SIII in ESM Appendix S5; Cluster ID 4; window length: 0.01 s); Territorial Advertisement (c, refers to Table SIII in ESM Appendix S5; Cluster ID 7; window length: 0.01 s); Group Cohesion Call (d, refers to Table SIII in ESM Appendix S5; Cluster ID 2; window length: 0.01 s); Alarm Call (e, refers to Table SIII in ESM Appendix S5; Cluster ID 1; window length: 0.01 s); Scream (f, refers to Table SIII in ESM Appendix S5; Cluster ID 6; window length: 0.01 s); Scream (g, refers to Table SIII in ESM Appendix S5; Cluster ID 10; window length: 0.01 s); Scream (h, refers to Table SIII in ESM Appendix S5; Cluster ID 5; window length: 0.01 s); Grunt (i, refers to Table SIII in ESM Appendix S5; Cluster ID 3; window length: 0.005 s); Grunt Hoot (j, refers to Table SIII in ESM Appendix S5; Cluster ID 8; window length: 0.01 s). A schematic representation of the polar diagram (Fig. 2) shows cluster and spectrogram position in the original figure.

Eulemur rubriventer

a

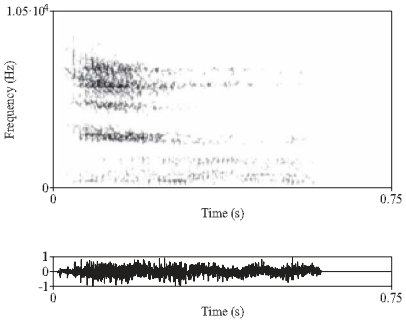
Hoot-Tonal Call

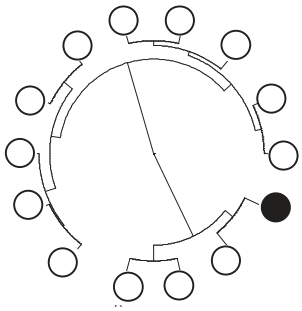




b

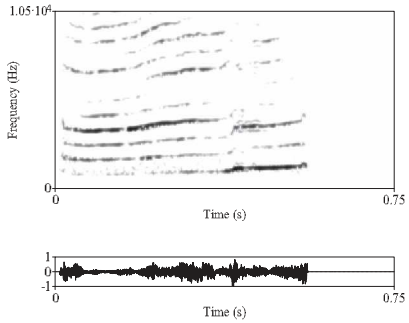
Long Grunt

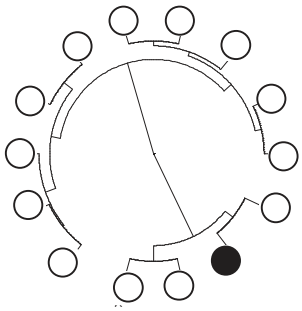




c

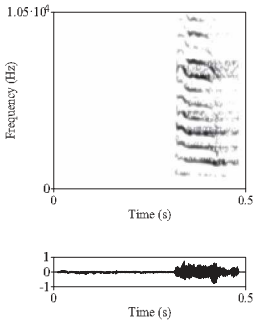
Tonal Call

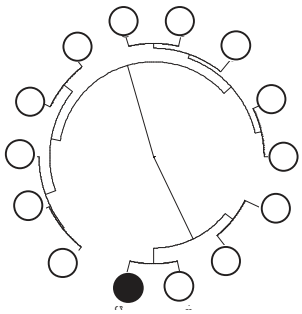




d

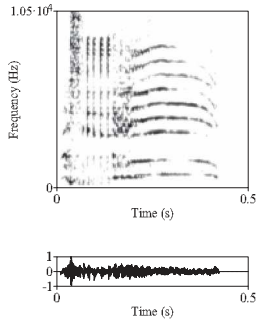
Hoot-Tonal Call

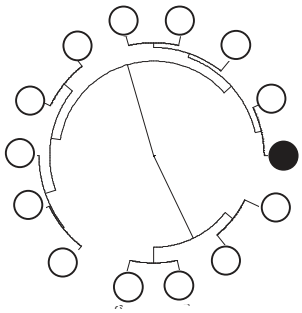




e

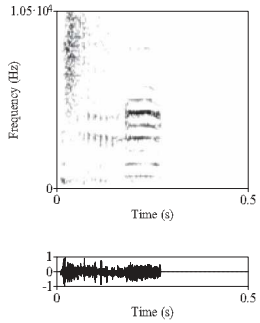
Snort-Grunt-Tonal Call

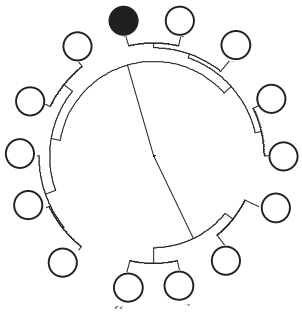




f

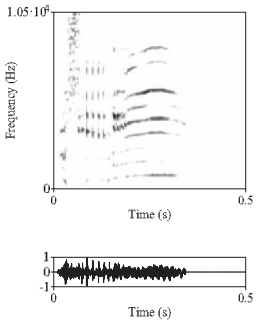
Snort-Grunt-Tonal Call

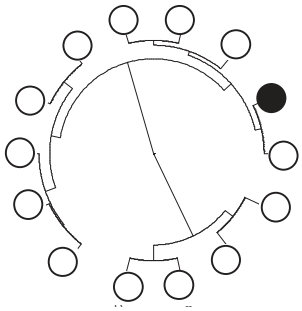




g

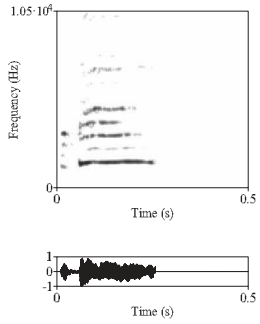
Snort-Grunt-Tonal Call

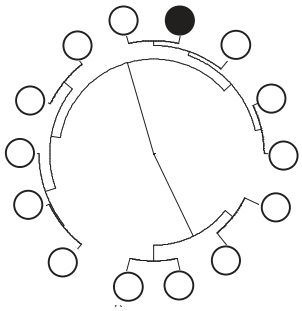




h

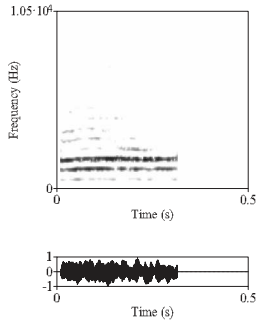
Tonal Call

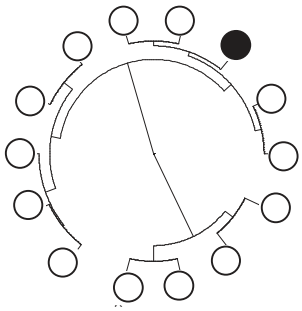




i

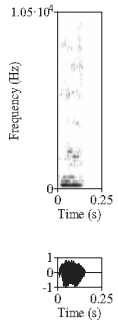
Tonal Call

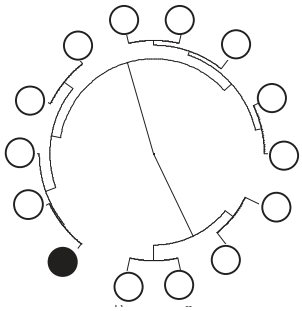




j

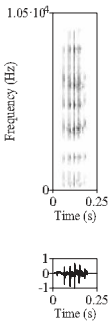
Hoot

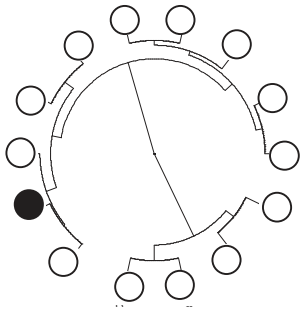




k

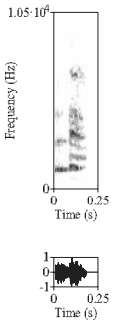
Grunt

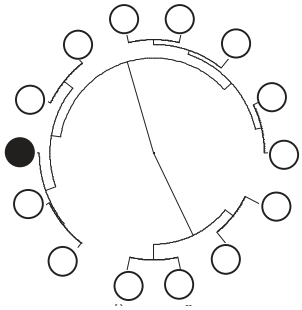




l

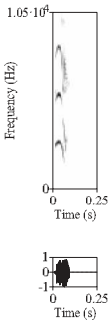
Gurgle

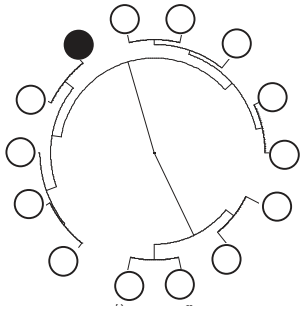




m

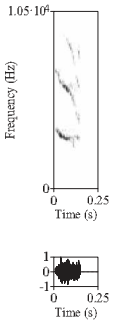
Scream





n

Scream



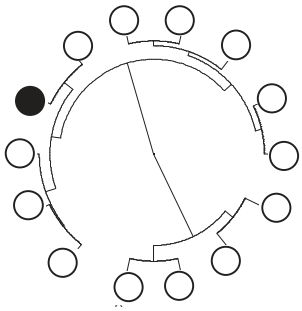
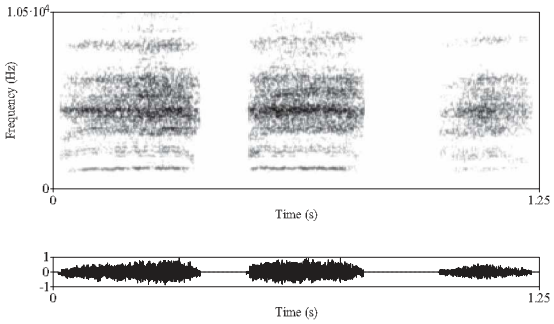


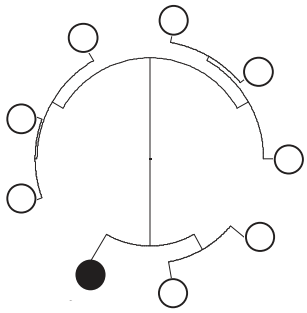
Fig. 18 Spectrograms and waveforms of the exemplars chosen from the vocalization of *Eulemur rubriventer* during the cluster analysis: Hoot-Tonal Call (a, refers to Table SIV in ESM Appendix S5; Cluster ID 8; window length: 0.01 s); Long Grunt (b, refers to Table SIV in ESM Appendix S5; Cluster ID 13; window length: 0.01 s); Tonal call (c, refers to Table SIV in ESM Appendix S5; Cluster ID 2; window length: 0.01 s); Hoot-Tonal Call (d, refers to Table SIV in ESM Appendix S5; Cluster ID 4; window length: 0.01 s); Snort-Grunt-Tonal Call (e, refers to Table SIV in ESM Appendix S5; Cluster ID 3; window length: 0.01 s); Snort-Grunt-Tonal Call (f, refers to Table SIV in ESM Appendix S5; Cluster ID 10; window length: 0.01 s); Snort-Grunt-Tonal Call (g, refers to Table SIV in ESM Appendix S5; Cluster ID 11; window length: 0.01 s); Tonal Call (h, refers to Table SIV in ESM Appendix S5; Cluster ID 7; window length: 0.01 s); Tonal Call (i, refers to Table SIV in ESM Appendix S5; Cluster ID 12; window length: 0.01 s); Hoot (j, refers to Table SIV in ESM Appendix S5; Cluster ID 6; window length: 0.01 s); Grunt (k, refers to Table SIV in ESM Appendix S5; Cluster ID 5; window length: 0.005 s); Gurgle (l, refers to Table SIV in ESM Appendix S5; Cluster ID 14; window length: 0.005 s); Scream (m, refers to Table SIV in ESM Appendix S5; Cluster ID 1; window length: 0.01 s); Scream (n, refers to Table SIV in ESM Appendix S5; Cluster ID 9; window length: 0.01 s). A schematic representation of the polar diagram (Fig. 3) shows cluster and spectrogram position in the original figure.

Eulemur mongoz

a

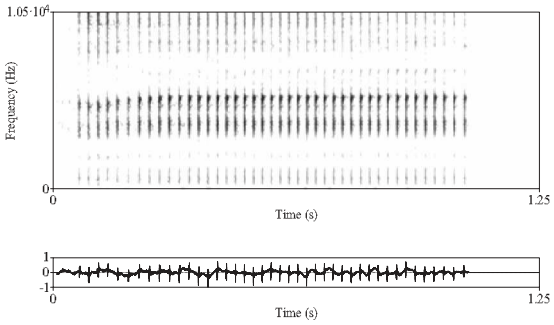
Group Cohesion Call

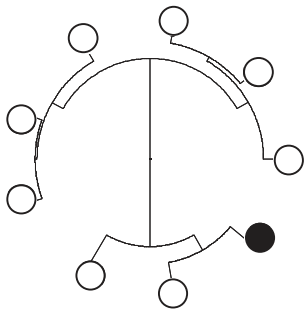




b

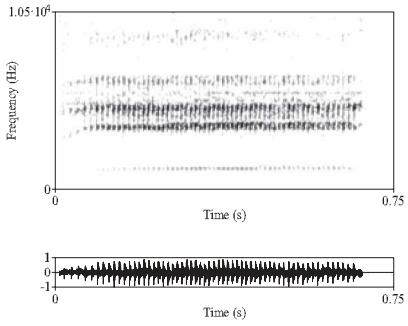
Long Grunt

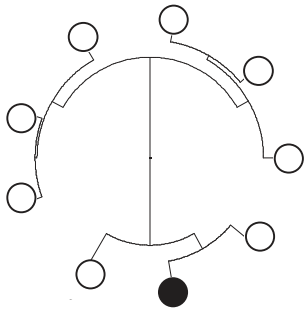




c

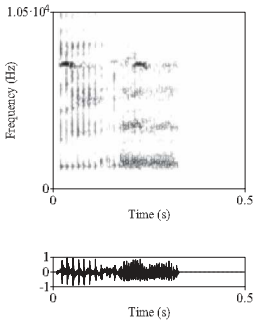
Alarm Long Grunt

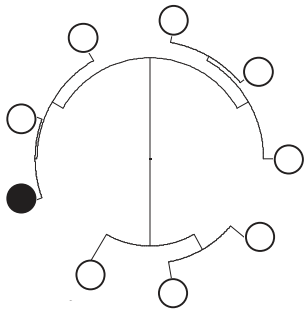




d

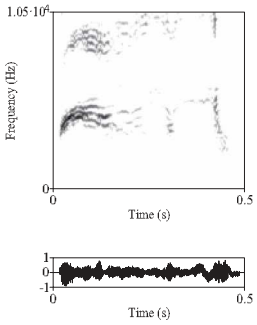
Grunt-Tonal Call

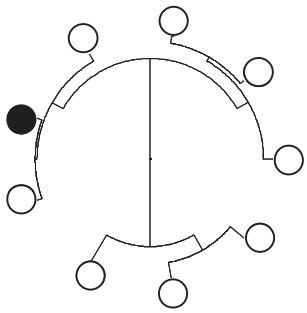




e

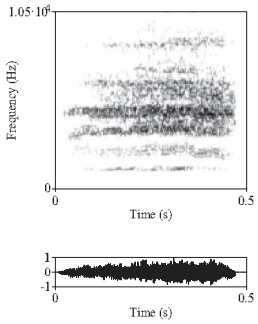
Scream

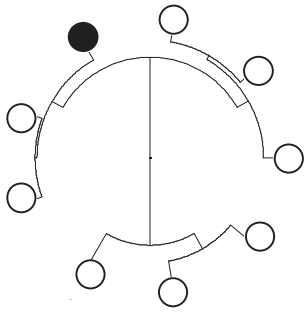




f

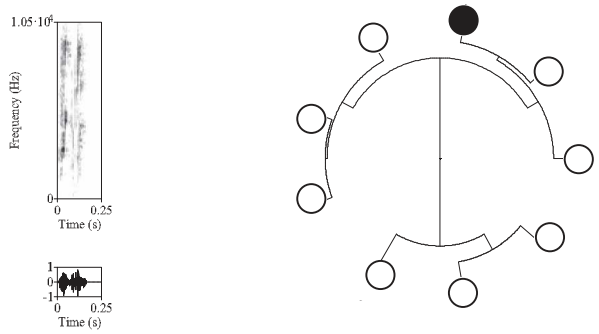
Territorial Advertisement Call





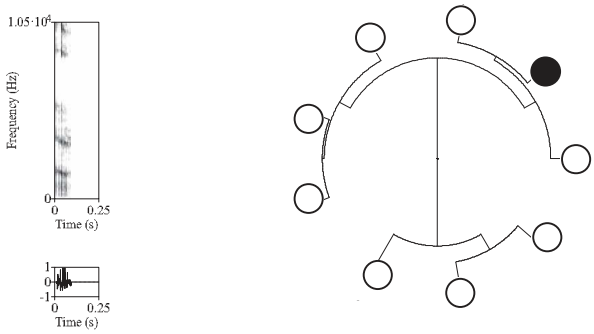
g

Click



h

Grunt



i

Grunted Hoot

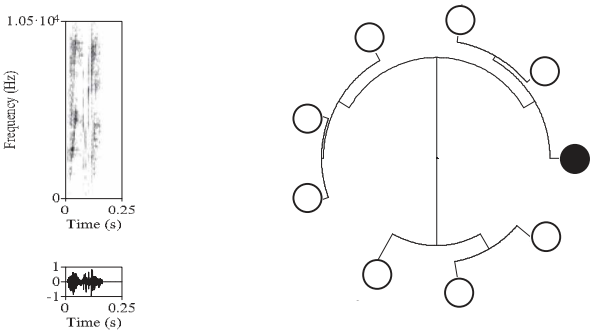
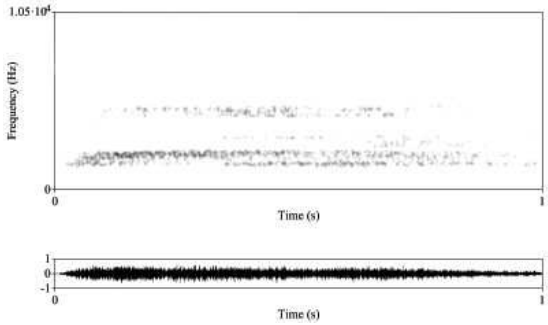


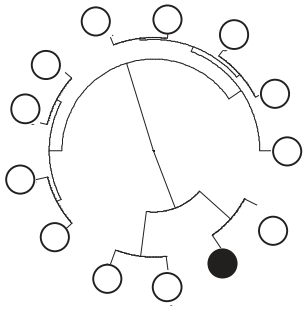
Fig. 19 Spectrograms and waveforms of the exemplars chosen from the vocalization of *Eulemur mongoz* during the cluster analysis: Cohesion Call (a, refers to Table SV in ESM Appendix S5; Cluster ID 1; window length: 0.01 s); Long Grunt (b, refers to Table SV in ESM Appendix S5; Cluster ID 5; window length: 0.01 s); Alarm Long Grunt call (c, refers to Table SV in ESM Appendix S5; Cluster ID 3; window length: 0.01 s); Grunt-Tonal Call (d, refers to Table SV in ESM Appendix S5; Cluster ID 7; window length: 0.01 s); Scream (e, refers to Table SV in ESM Appendix S5; Cluster ID 4; window length: 0.01 s); Territorial Advertisement Call (f, refers to Table SV in ESM Appendix S5; Cluster ID 8; window length: 0.01 s); Click (g, refers to Table SV in ESM Appendix S5; Cluster ID 9; window length: 0.005 s); Grunt (h, refers to Table SV in ESM Appendix S5; Cluster ID 6; window length: 0.005 s); Grunted Hoot (i, refers to Table SV in ESM Appendix S5; Cluster ID 2; window length: 0.005 s). A schematic representation of the polar diagram (Fig. 4) shows cluster and spectrogram position in the original figure.

Eulemur coronatus

a

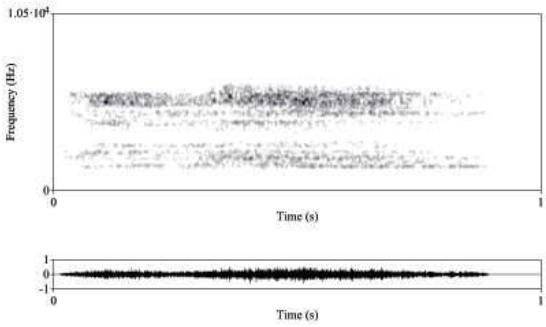
Alarm Call

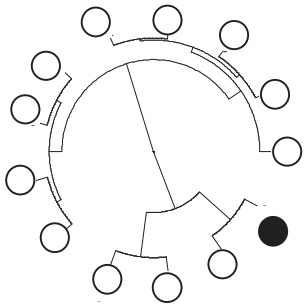




b

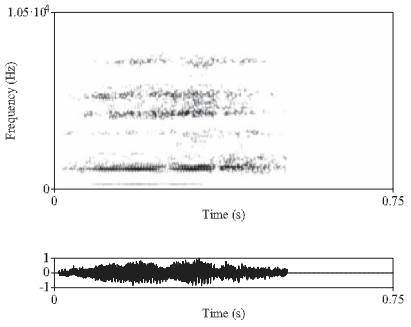
Alarm Call

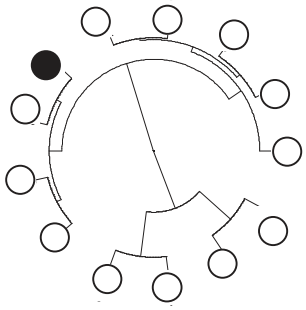




c

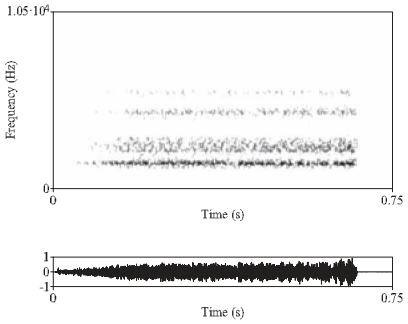
Alarm Call

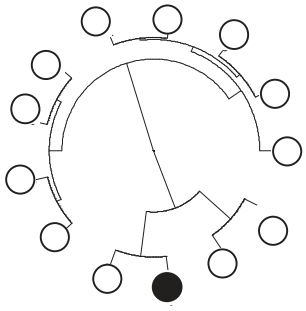




d

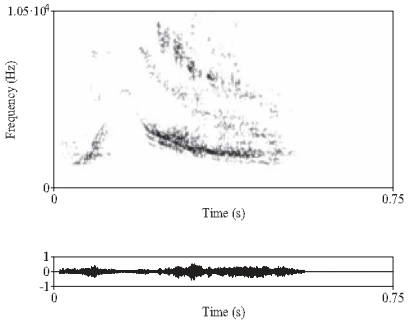
Alarm Call

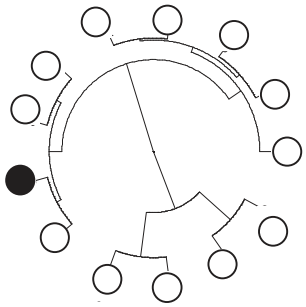




e

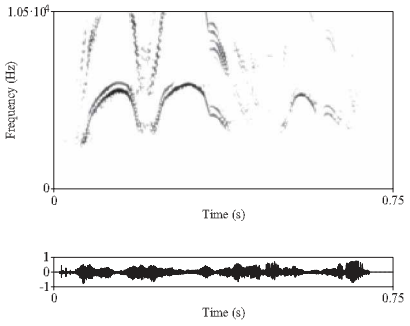
Scream

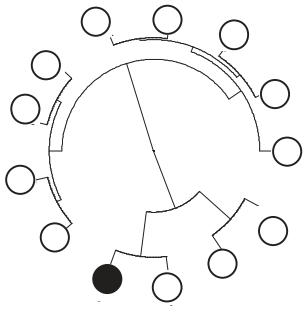




f

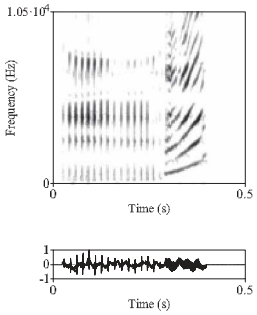
Scream

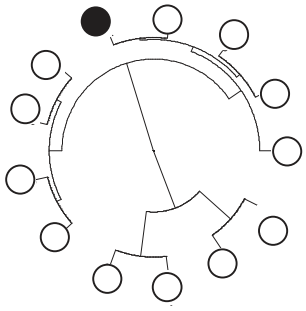




g

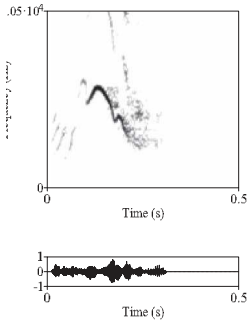
Grunt-Tonal Call

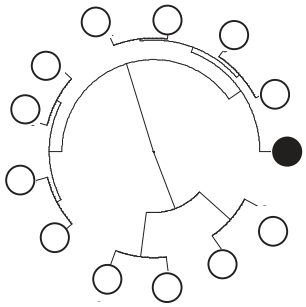




h

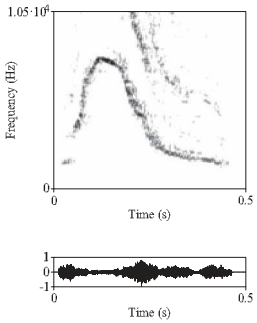
Scream

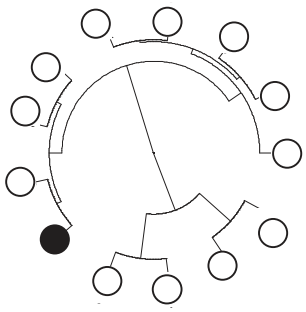




i

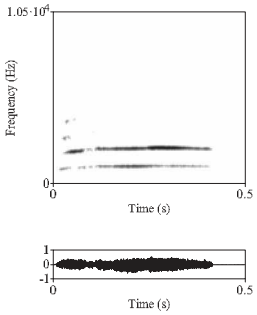
Scream

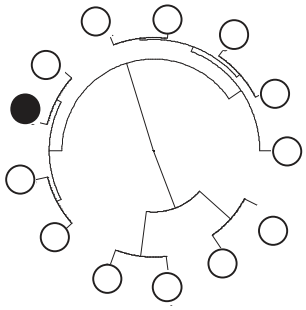




j

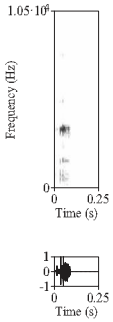
Tonal Call

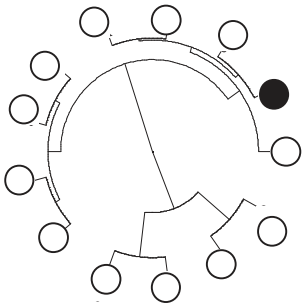




k

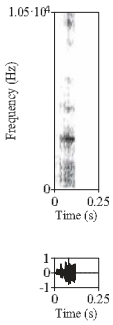
Grunt

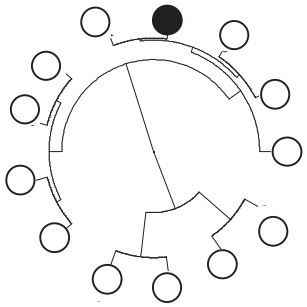




l

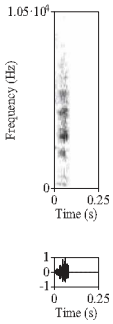
Grunt





m

Grunt



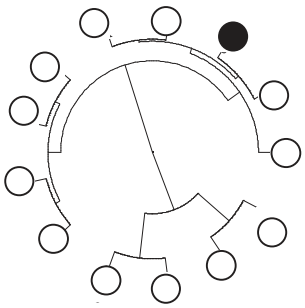
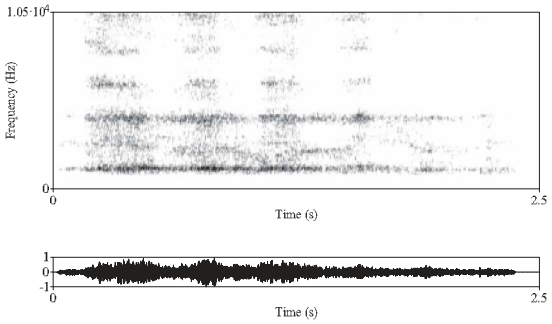


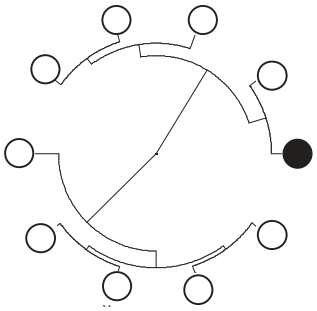
Fig. 20 Spectrograms and waveforms of the exemplars chosen from the vocalization of *Eulemur coronatus* during the cluster analysis: Alarm Call (a, refers to Table SVI in ESM Appendix S5; Cluster ID 3; window length: 0.01 s); Alarm Call (b, refers to Table SVI in ESM Appendix S5; Cluster ID 5; window length: 0.01 s); Alarm Call (c, refers to Table SVI in ESM Appendix S5; Cluster ID 2; window length: 0.01 s); Alarm Call (d, refers to Table SVI in ESM Appendix S5; Cluster ID 4; window length: 0.01 s); Scream (e, refers to Table SVI in ESM Appendix S5; Cluster ID 8; window length: 0.01 s); Scream (f, refers to Table SVI in ESM Appendix S5; Cluster ID 1; window length: 0.01 s); Grunt-Tonal Call (g, refers to Table SVI in ESM Appendix S5; Cluster ID 11; window length: 0.01 s); Scream (h, refers to Table SVI in ESM Appendix S5; Cluster ID 13; window length: 0.01 s); Scream (i, refers to Table SVI in ESM Appendix S5; Cluster ID 10; window length: 0.01 s); Tonal Call (j, refers to Table SVI in ESM Appendix S5; Cluster ID 6; window length: 0.01 s); Grunt (k, refers to Table SVI in ESM Appendix S5; Cluster ID 7; window length: 0.005 s); Grunt (l, refers to Table SVI in ESM Appendix S5; Cluster ID 9; window length: 0.005 s); Grunt (m, refers to Table SVI in ESM Appendix S5; Cluster ID 12; window length: 0.005 s). A schematic representation of the polar diagram (Fig. 5) shows cluster and spectrogram position in the original figure.

Eulemur flavifrons

a

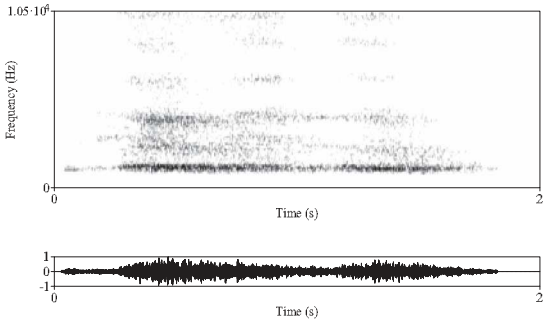
Alarm Call

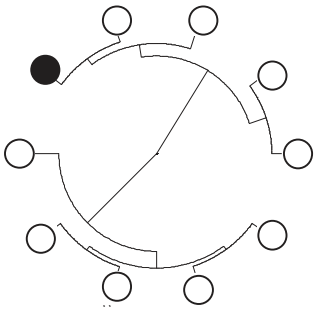




b

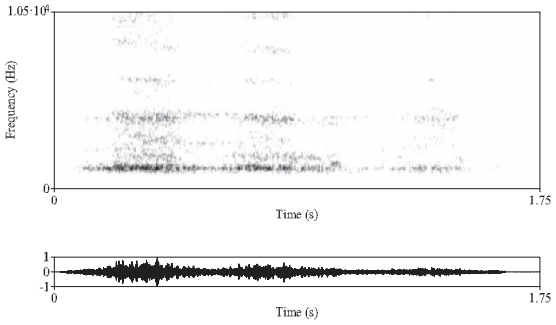
Alarm Call

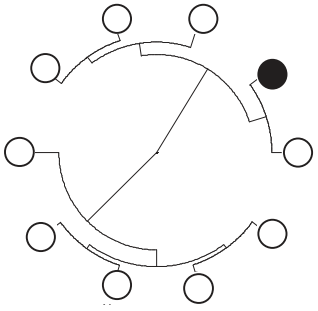




c

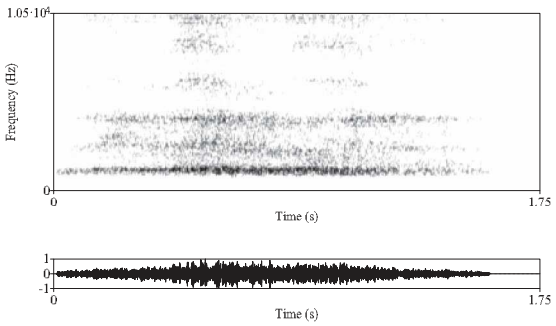
Alarm Call

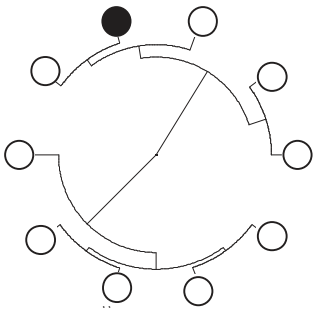




d

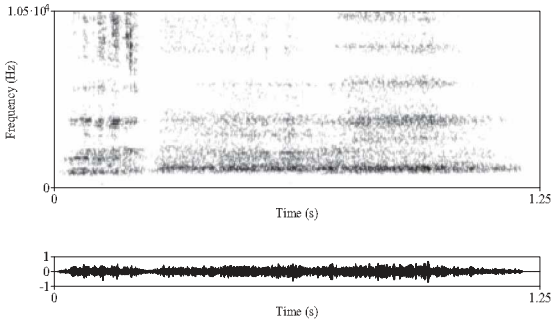
Alarm Call

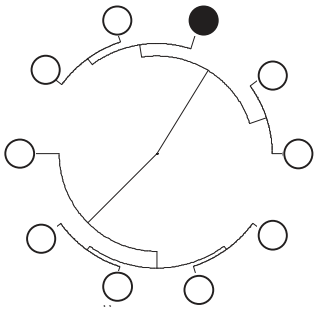




e

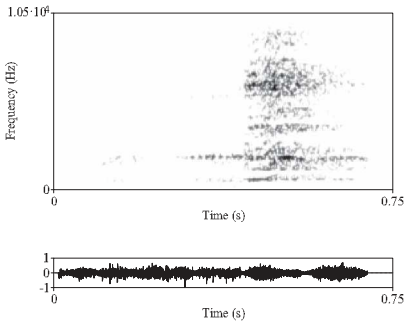
Alarm Call

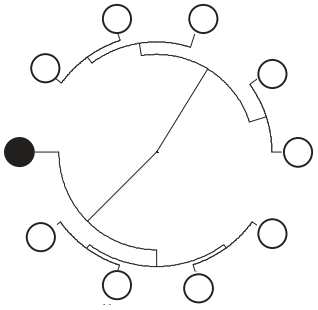




f

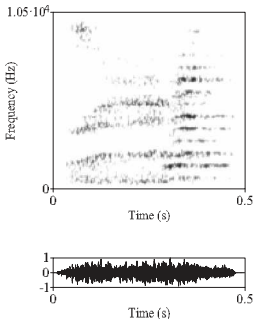
Long Grunt-Tonal Call

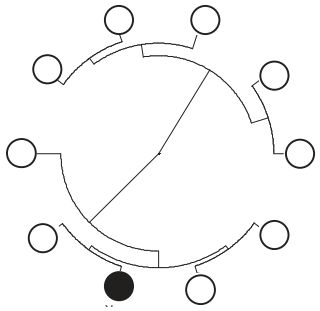




g

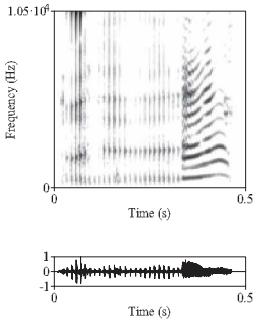
Grunt-Tonal Call

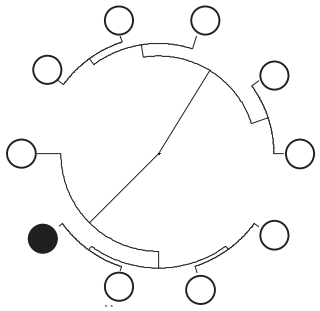




h

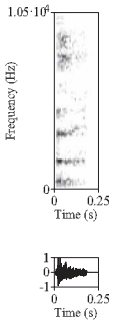
Long Grunt-Tonal Call

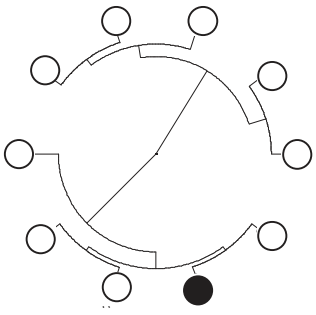




i

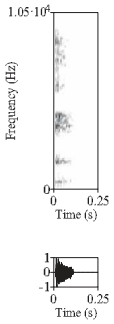
Grunted Hoot





j

Grunted Hoot



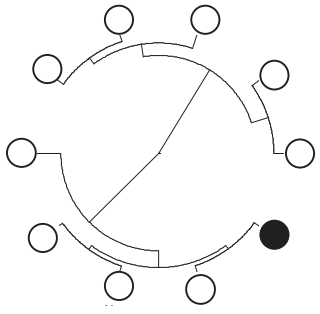
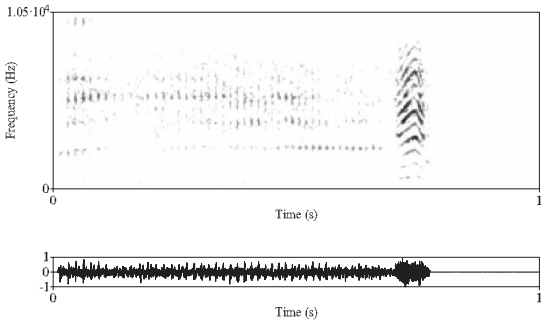


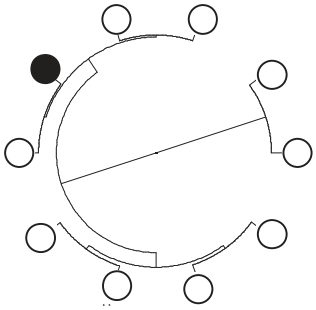
Fig. 21 Spectrograms and waveforms of the exemplars chosen from the vocalization of *Eulemur flavifrons* during the cluster analysis: Alarm Call (a, refers to Table SVII in ESM Appendix S5; Cluster ID 1; window length: 0.01 s); Alarm Call (b, refers to Table SVII in ESM Appendix S5; Cluster ID 8; window length: 0.01 s); Alarm Call (c, refers to Table SVII in ESM Appendix S5; Cluster ID 3; window length: 0.01 s); Alarm Call (d, refers to Table SVII in ESM Appendix S5; Cluster ID 7; window length: 0.01 s); Alarm Call (e, refers to Table SVII in ESM Appendix S5; Cluster ID 10; window length: 0.01 s); Long Grunt-Tonal Call (f, refers to Table SVII in ESM Appendix S5; Cluster ID 5; window length: 0.01 s); Grunt-Tonal Call (g, refers to Table SVII in ESM Appendix S5; Cluster ID 9; window length: 0.01 s); Long Grunt-Tonal Call (h, refers to Table SVII in ESM Appendix S5; Cluster ID 6; window length: 0.01 s); Grunted Hoot (i, refers to Table SVII in ESM Appendix S5; Cluster ID 2; window length: 0.005 s); Grunted Hoot (j, refers to Table SVII in ESM Appendix S5; Cluster ID 4; window length: 0.005 s). A schematic representation of the polar diagram (Fig. 6) shows cluster and spectrogram position in the original figure.

Eulemur macaco

a

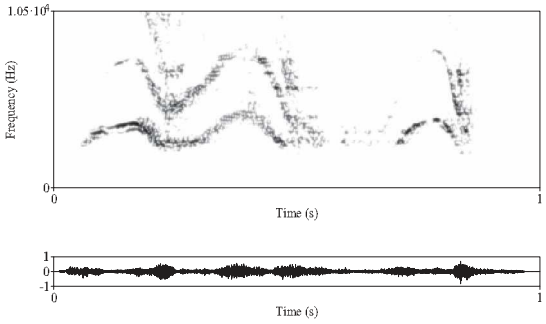
Long Grunt-Tonal Call

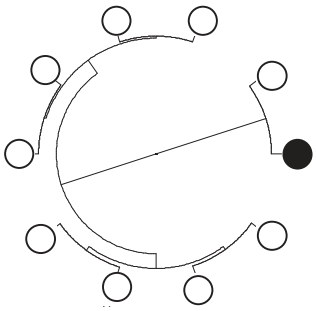




b

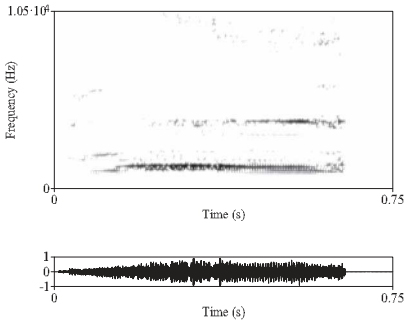
Scream

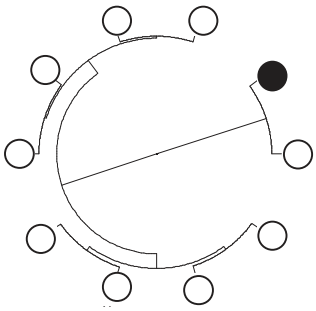




c

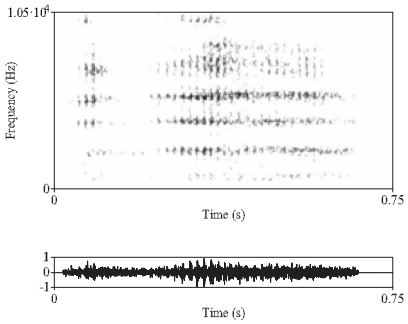
Alarm Call

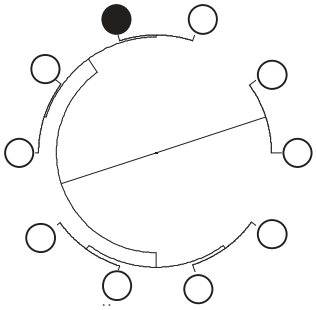




d

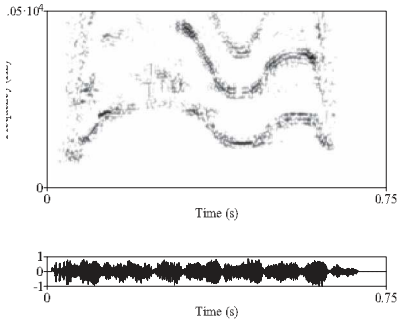
Long Grunt-Tonal Call

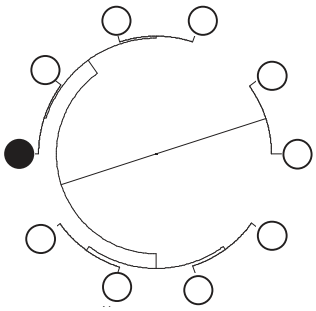




e

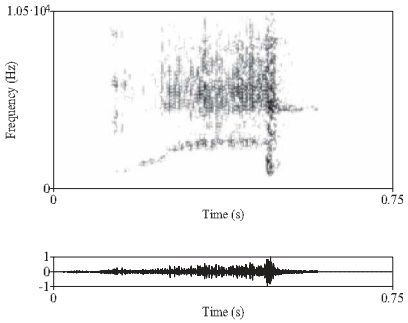
Scream

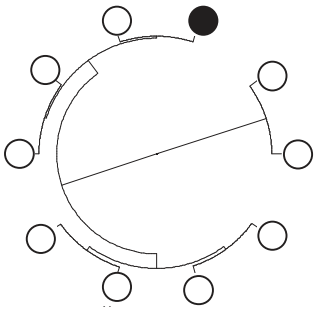




f

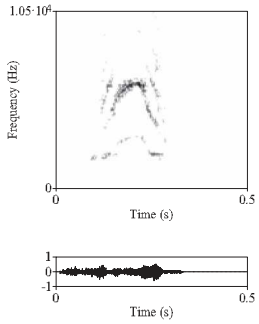
Scream

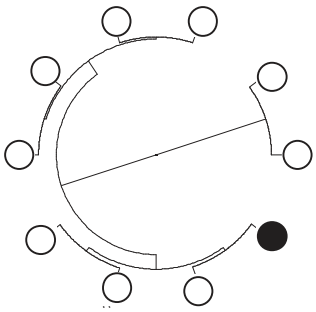




g

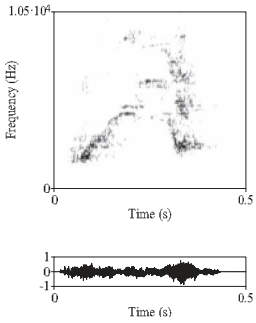
Scream

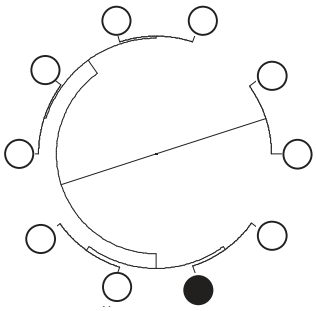




h

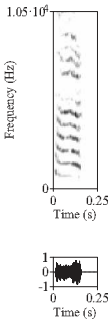
Scream

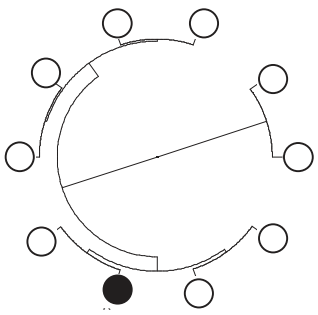




i

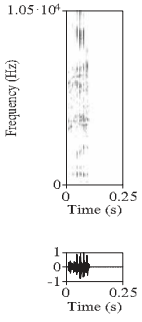
Tonal Call





j

Grunt



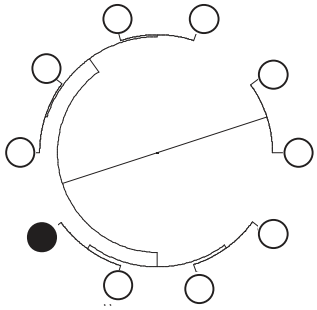


Fig. 22 Spectrograms and waveforms of the exemplars chosen from the vocalization of *Eulemur macaco* during the cluster analysis: Long Grunt-Tonal Call (a, refers to Table SVIII in ESM Appendix S5; Cluster ID 7; window length: 0.01 s); Scream (b, refers to Table SVIII in ESM Appendix S5; Cluster ID 1; window length: 0.01 s); Alarm Call (c, refers to Table SVIII in ESM Appendix S5; Cluster ID 8; window length: 0.01 s); Long Grunt Tonal Call (d, refers to Table SVIII in ESM Appendix S5; Cluster ID 9; window length: 0.01 s); Scream (e, refers to Table SVIII in ESM Appendix S5; Cluster ID 10; window length: 0.01 s); Scream (f, refers to Table SVIII in ESM Appendix S5; Cluster ID 4; window length: 0.01 s); Scream (g, refers to Table SVIII in ESM Appendix S5; Cluster ID 5; window length: 0.01 s); Scream (h, refers to Table SVIII in ESM Appendix S5; Cluster ID 2; window length: 0.01 s); Tonal Call (i, refers to Table SVIII in ESM Appendix S5; Cluster ID 6; window length: 0.01 s); Grunt (j, refers to Table SVIII in ESM Appendix S5; Cluster ID 3; window length: 0.005 s). A schematic representation of the polar diagram (Fig. 7) shows cluster and spectrogram position in the original figure.